Inferential Statistics

Key terms **KSBs: K13, K54, S11, S52, B1**

sampling, bias, central limit theorem, confidence intervals

One of the most powerful aspects of statistics is the way it allows us to make inferences about larger populations using data from smaller samples.

Definition 4.1 - Statistics and Parameters

A **statistic** is a quantity calculated from a sample. A population has **parameters**, such as the mean, which can be estimated from a sample using a statistic.

In general, when we are using a sample to make estimates, the larger samples we can get, the better our approximations will be.

Example 4.1. *Our population consists of the weights of* 1000 eggs *laid in a week by a flock of hens. We cannot weigh all the eggs, so we choose to take a* Simple random sample *and weigh those.*

Suppose our first sample consists of 5 eggs with the following masses (in grams):

$$58.6 \quad 61.0 \quad 63.0 \quad 64.0 \quad 66.8$$

The mean of this sample is $\bar{x}_1 = 62.7$. *Suppose we take another sample and this time it is*

$$52.8 \quad 57.1 \quad 58.8 \quad 59.9 \quad 62.5$$

The mean of this sample is $\bar{x}_2 = 58.2$

It is very likely that the population mean is neither of these values. *These samples are* small *so may not give an accurate reflection of the whole population. Depending on how the sample of eggs was chosen, they may not be* representative.

Exercise 4.1. *The data below is a random sample of the number of goals scored in 10 Premier league football matches taken from the first 3 weeks of the 2020/21 season.*

$$0 \quad 3 \quad 1 \quad 0 \quad 6 \quad 4 \quad 1 \quad 1 \quad 3 \quad 2$$

(a) Use the sample to estimate the mean number of goals likely to be scored throughout the season.
(b) Make at least 2 comments on the accuracy of the estimate.

Sampling icon by Naba A'la Lail from the Noun Project

Method	Description	Benefits	Weaknesses	Use case
Census	Survey every member of a population	Gets you the full data set	Often impractical to implement	Generally only for small populations
Simple Random Sample	Pick members of the population at random, without replacement	Free from bias	May not be representative, if the sample is small	If you can guarantee those you select will participate
Unrestricted Random Sampling	Pick members of the population at random, with replacement	Free from bias	May not be representative, if the sample is small	If your population is 'visits' or instances of something
Cluster Sampling	Split population into clusters and take a random sample from each	Allows you to compare between clusters	May not be representative, if the sample is small	If your data falls naturally into clusters (e.g staff at branches)
Proportional Stratified Sampling	Consider the population split into strata (by e.g. gender, class etc) and take a sample from each proportional to the strata's proportion of the population	Gives a better estimate for the population overall	Need to know about the population to establish strata; gives less info about smaller groups	When you need an estimate for the overall population
Disproportional Stratified Sampling	Consider the population split into strata (by e.g. gender, class etc) and take a sample from each, of any size	Allows you to collect the same number from each stratum, or more from strata with greater variability	Less good estimate of the whole population	When you would like to study underrepresented groups in a population
Systematic Sampling	Taking samples at regular intervals in a population	Gives representative results	Increased risk of data manipulation; needs natural randomness	When your population has a natural ordering (e.g. customers arriving)
Judgemental or Purposive Sampling	You or an expert chooses members of the population you think will be most useful to the investigation	Gets you more useful data; easy to implement	Extremely prone to bias	When you have expert knowledge
Convenience Sampling	You collect data from the most accessible members of the population	Easy to implement	Not a representative sample	For initial data that you don't plan to generalise
Voluntary Response Sampling	You put out a survey to the whole population and some members respond	Less work to implement	Often biased in favour of strong opinions, not representative	When your resources are low, or you just want to find out what strong opinions are
Snowball or Network Sampling	You recruit a small number of participants and they recommend others	Allows you to access specific communities	Not representative of the whole population; can be difficult	When the population you want to study is hard to access

Exercise 4.2.

(a) For each of the methods described below, identify the type of sampling being used.

(i) You assign a number to every employee in the company database from 1 to 1000, and use a random number generator to select 100 numbers.

(ii) All employees of the company are listed in alphabetical order. From the first 10 numbers, you randomly select a starting point: 6. From 6 onwards, every 10th person on the list is selected (6, 16, 26, 36, and so on), and you end up with a sample of 100 people.

(iii) You collect data from everyone who sits on the desks near you in your office.

(iv) You want to know more about the opinions and experiences of employees with a disability in the company, so you contact a sample of employees with support needs.

(v) You send out the survey to all staff at your company and many staff members decide to complete it.

(vi) The company has 800 female employees and 200 male employees. You select 80 women and 20 men, to get a sample of 100 people.

(vii) You want to find out about the opinions of people who like playing board games at your company. You know someone in your office who plays regularly with other staff members, so you ask them to suggest others you could interview.

(viii) The company has offices in 10 cities across the country (all with roughly the same number of employees in similar roles). You select 3 offices and choose a random sample from each.

(b) A large number of spectators attend a football match at a ground in England. One of the stands contains 1390 seats, numbered from 1 to 1390.

(i) Describe how random numbers could be used to select a random sample of 80 of these seats.

(ii) The statistician proposes to ask the spectators occupying the 80 seats selected in part (i) to complete a questionnaire. Suggest two practical difficulties with this.

(c) The statistician wishes to investigate the amount of interest taken in sport by the population of the UK. They plan for interviewers to be stationed outside each exit from the ground, to interview every 100th person leaving after the match.

(i) State the name given to this type of sampling.

(ii) Give one reason why the results of the investigation suggested might be biased.

(iii) Suggest two other practical difficulties that might be encountered in carrying out these interviews.

4.1 Sampling and Bias

Some of the sampling methods we have discussed introduce obvious forms of bias - the sample may be biased due to not being representative of the whole population, or because the method of selection has skewed the types of observations we make in favour of particular outcomes.

Other types of bias can also affect statistical data gathering, including in ways you might not expect.

- **Confirmation bias** is only gathering data which supports your belief. This will mainly occur in sampling which involves you making decisions, like judgemental sampling or convenience sampling - e.g. only asking people who agree with you, or assuming because something has been effective in your own experience that it is better.

- **Self-selection bias** is when certain types of subjects are more likely to include themselves in the sample. This can happen with voluntary surveys or if a convenience sample is taken e.g. in a particular restaurant, where people are statistically more likely to be fans of that restaurant.

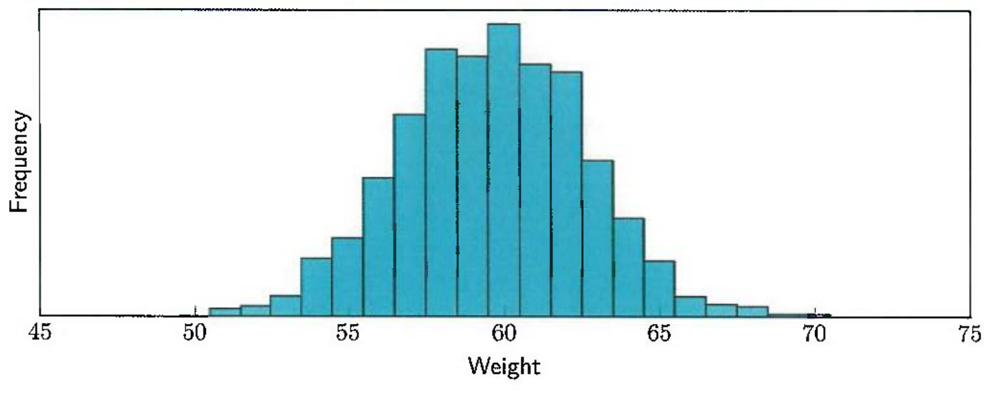
- **Survival bias** only captures living & survived subjects, while ones no longer around are not accounted for. This could be because of your sampling method, or an unforeseen factor influencing subjects' ability to participate.

The mathematical techniques we use to analyse data will not be able to tell when the data you give it is biased or unrepresentative, so it's important to consider factors like this before feeding data in - especially when it will be used to make decisions or train models.

4.2 The Central Limit Theorem

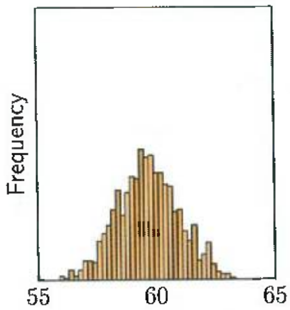
When we take a sample, it may not be representative of the whole population. But what if we were to take many different random samples from the same population and compare them?

Example 4.2. *Consider our eggs example from earlier (Example 4.1). The population is shown in this histogram. It appears to have a* normal distribution *which is what we might expect for this kind of data. The data has a mean of* 59.66 *and a standard deviation of* 3.03.



From the whole population of 1000 eggs, we could take 1000 different random samples of size 5 and look at their means.

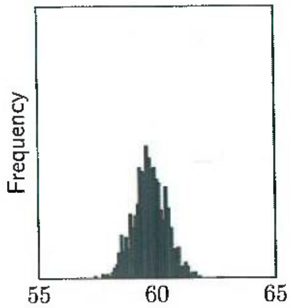
The means of the samples also appear to follow a normal distribution. The mean of the sample means is 59.67 and the standard deviation is 1.35. This is the same mean as the original data, but a smaller S.D.



Sample means ($n = 5$)

If we now take 1000 samples of size 20, and plot their means, we get a slightly different distribution.

This is still distributed normally with a mean of 59.69 but the standard deviation is now 0.68



Sample means ($n = 20$)

This might seem unsurprising, given that the original data was normally distributed. But in fact, this same behaviour is observed for any kind of data, even if the distribution is uniform, or any other non-uniform distributions.

This is because of a result in statistics called the **Central Limit Theorem**, and it allows us to apply the tools we have to study the normal distribution to analyse any kind of data set.

Theorem 4.1 - Central Limit Theorem

If we take random samples of size n from a population with mean μ and standard deviation σ and calculate the mean of each sample, the distribution of sample means will have the following properties:

- The mean of the distribution will be equal to the population mean μ.
- If the population is normally distributed, then the sample means will be normally distributed.
- If the population is not normal, but the sample size is greater than 30 the sample means will roughly follow a normal distribution
- The standard deviation of the distribution of sample means, also called the **standard error,** is given by $\frac{\sigma}{\sqrt{n}}$ (variance is $\frac{\sigma^2}{n}$)
- We write

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$$

Example 4.3. *For the eggs data, the standard deviation of the original data was 3.03, and the standard deviation of the means of samples of size 5 was 1.35.* ~~Then~~ *For samples of size 20, the standard deviation was 0.68*

$$\frac{3.03}{\sqrt{5}} = \frac{3.03}{2.22} = 1.35.$$

Similarly,

$$\frac{3.03}{\sqrt{20}} = \frac{3.30}{4.47} = 0.67$$

The reason this value is called the standard error is that we can use it to measure the accuracy of an estimate. The central limit theorem tells us about the distribution of sample means, but in general, you will only have one sample of data to study.

The standard error tells us about how likely the true population mean is to be close to the mean of our sample, and the smaller it is, the closer we can pin down the location of μ.

Example 4.4. *A test was given to 120 students. The marks were normally distributed, with a* S.D of 20. *The following marks are from a random sample of 12 students:*

35 23 17 38 20 25 29 32 28 31 33 24

*The mean of this sample can be found by adding together the values in the sample and dividing by the size of the sample (*12*).*

$$\frac{35+23+17+38+20+25+29+32+28+31+33+24}{12} = \frac{335}{12} = 27.91$$

Since the data is normally distributed, even though the sample size is less than 30 *this can be used as an estimate for* the population mean.

The standard error for this sample can be found by dividing the SD of the whole population (20) by the square root of the sample size:

$$\frac{\sigma}{\sqrt{12}} = \frac{20}{\sqrt{12}} = \frac{20}{3.46} = 5.77.$$

If we wanted to reduce the standard error to half its current value, we could increase the size of the samples. *With samples of size* 48, *we would have* $\frac{20}{\sqrt{48}} = \frac{20}{6.93} = 2.88$ *This is* half the previous standard error.

In general, if we want to halve the standard error (divide it by 2), we need to multiply the sample size by $2^2 = 4$ To make the standard error a third of its previous size (divide it by 3), we need to multiply the sample size by $3^2 = 9$.

Example 4.5. *Suppose there is a large bag of coins; 10% are 10p; 75% are 20p and the remaining 15% are 50p.*

Then if X is the value of a coin picked out of the bag at random, X is a discrete random variable with the given probability distribution.

x	10	20	50
$P(X = x)$	0.1	0.75	0.15

Let $\bar{X}$ be the mean value of a coin from a random sample of size 50. By the Central Limit Theorem, since $n > 30$, the distribution of $\bar{X}$ is normal - even though X is a discrete random variable. Because $\bar{X}$ is the distribution of sample means, there are a lot more than 3 possible values - enough that we may treat it as continuous.

We can calculate the mean and standard deviation of X, as we have previously:

$$\mu = \sum x P(X=x) = (10 \times 0.1) + (20 \times 0.75) + (50 \times 0.15) = 23.5$$

$$\sigma^2 = \sum (x-\mu)^2 P(X=x) = ((10-23.5)^2 \times 0.1) + ((20-23.5)^2 \times 0.75) + ((50-23.5)^2 \times 0.15)$$

$$= 18.225 + 9.1875 + 105.3375 = 132.75 \quad \text{so } \sigma = 36.77.$$

This means that, even though X is a discrete random variable, by the Central Limit Theorem:

$$\bar{X} \sim N\left(23.5, \frac{132.75}{50}\right)$$

The normal approximation to the binomial distribution is actually a special case of the Central Limit Theorem.

Exercise 4.3. *(a) Let $X \sim N(34, 6^2)$. Let $\bar{X}$ be the sampling distribution of the mean of a random sample of 12 observations. Find the standard error of the mean.*

(b) Let $X \sim N(23.4, 25)$. Let $\bar{X}$ be the sampling distribution of the mean of a random sample of 26 observations.

(i) Find the standard error of the mean.

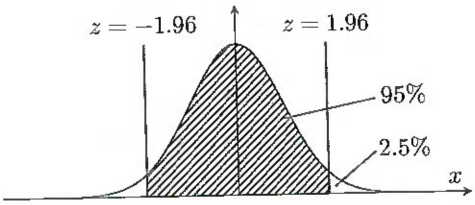
(ii) Write down the distribution of $\bar{X}$.

4.3 Confidence intervals

Definition 4.2 - Confidence Interval

For a sample of a population, where the sample mean is $\bar{x}$ and the population mean is μ the **confidence interval** is a range of values where the value of μ is likely to be, centred around $\bar{x}$. We can use the fact that the sample means are normally distributed to calculate the size of this interval, by finding the z-score for the level of confidence we want.

Suppose we would like to know with 95% confidence where the population mean lies. This means we need to find the line with probability 97.5% (since a 95% section in the centre will end at 97.5%). Looking at our table for the normal distribution, 97.5% $\Rightarrow$ z-score of 1.96.



Let a be the value we are looking for, which is the size of the interval. Recall that the z-score is calculated as $z = \frac{x-\mu}{\sigma}$ The distribution of sample means is given by $\bar{X} \sim N(\mu, \frac{\sigma^2}{n})$ so we have $\mu = \mu$, and σ equal to the standard error (SE).

$$1.96 = \frac{a-\mu}{SE} \Rightarrow a = \mu + (1.96 \times SE).$$

So the interval where the population mean might be extends a distance of a to the right of the sample mean. But if $\mu + 1.96\,SE \le \bar{x}$ we can subtract $\bar{x}$ and μ on both sides to get

$$-\bar{x} + 1.96\,SE \le -\mu \quad \text{or} \quad \mu \le \bar{x} - 1.96\,SE.$$

Since the distribution is symmetrical, this inequality also holds the other way round, So we have

$$\bar{x} - 1.96\,SE \le \mu \le \bar{x} + 1.96\,SE.$$

Definition 4.3 - 95% Confidence Interval for the Population Mean

For a sample of size n, the interval

$$\left(\bar{x} - 1.96\frac{\sigma}{\sqrt{n}},\ \bar{x} + 1.96\frac{\sigma}{\sqrt{n}}\right)$$

is the 95% confidence interval for the population mean.

Example 4.6. *The eggs data had a mean of $\mu = 59.66$ and a standard deviation of $\sigma = 3.03$. If we took a sample of size 5 which has* a mean of $\bar{x} = 60.2$ *we would find* the interval where we are 95% confident μ is *to be:*

$$\left(60.2 - 1.96 \times \frac{3.03}{\sqrt{5}},\ 60.2 + 1.96 \times \frac{3.03}{\sqrt{5}}\right) = (57.56, 62.84)$$

Reducing the size of the standard error, by increasing the sample size, allows us to narrow the confidence interval, so we can be more certain of the location of the population mean.

Exercise 4.4.

(a) *Write down the z-score required for*
   (i) *A 90% confidence interval* (ii) *A 98% confidence interval* (iii) *A 99% confidence interval*

(b) *Let $X \sim N(\mu, 12^2)$. Find a 95% confidence interval for μ based on a sample of size 15 with mean 24.*

(c) *Let $X \sim N(\mu, 24^2)$. Find a 98% confidence interval for μ based on a sample of size 30 with mean 94.*

(d) *Let $X \sim N(\mu, 2.35^2)$. Find a 90% confidence interval for μ based on a sample of size 10 with mean 165.*

This all relies on us knowing the population standard deviation. If we don't know this, or the size of the samples is < 30 and the underlying distribution is not known to be normal, we need to instead rely on the t-distribution which we will see more about later.